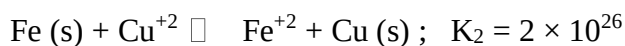
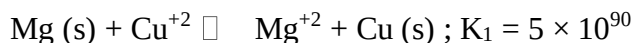


Subjective-Assignment (Equilibrium)

1. Both metals Mg and Fe can reduce copper from a solution having Cu^{+2} ion according to equilibria.



Which metal will remove cupric ion from the solution to a greater extent.

2. For the reaction, $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$; $K = 0.059$ atm at 1000 K. 1 g of CaCO_3 is placed in a 10 litre container at 1000 K to reach the equilibrium. Calculate the mass of CaCO_3 left at equilibrium.

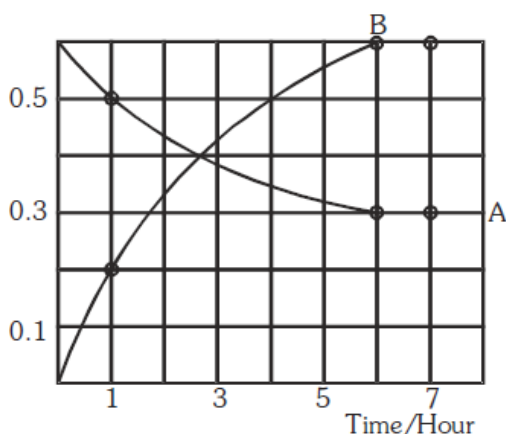
3. For the reaction, $[\text{Ag}(\text{CN})_2]^- \rightleftharpoons \text{Ag}^+ + 2\text{CN}^-$, the equilibrium constant, K_c at 27°C is 4.0×10^{-19} . To find the silver ion concentration in a solution which is originally 0.10 M in KCN and 0.03 M in AgNO_3 .

4. K_p for the reaction $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ is 0.66 at 46°C . Calculate the percent dissociation of N_2O_4 at 46°C and a total pressure of 0.5 atm. Also calculate the partial pressure of N_2O_4 and NO_2 at equilibrium.

4. The progress of the reaction

$\text{A} \rightleftharpoons n\text{B}$ with time, is presented in figure, Determine

(i) the value of n.



(ii) the equilibrium constant K.

(iii) the initial rate of conversion of A.

6. The decomposition of solid ammonium carbamate, $(\text{NH}_4)(\text{NH}_2\text{CO}_2)$, to gaseous ammonia and carbon dioxide is an endothermic reaction.



- (1) When solid $(\text{NH}_4)(\text{NH}_2\text{CO}_2)$ is introduced into an evacuated flask at 25°C , the total pressure of gas at equilibrium is 0.116 atm. What is the value of K_p at 25°C ?
- (2) Given that the decomposition reaction is at equilibrium, how would the following changes affect the total quantity of NH_3 in the flask once equilibrium is re-established?
- Adding CO_2
 - Adding $(\text{NH}_4)(\text{NH}_2\text{CO}_2)$
 - Removing CO_2
 - Increasing the total volume
 - Adding neon
 - Increasing the temperature
7. 500 ml of 10^{-5} M NaOH is mixed with 500 ml of 2.5×10^{-5} M of $\text{Ba}(\text{OH})_2$. To the resulting solution 99 L water is added. Calculate pH.
8. Calculate pH of 10^{-7} M of NaOH solution
9. What will be the resultant pH when 150 mL of an aqueous solution of HCl (pH = 2.0) is mixed with 350 mL of an aqueous solution of NaOH (pH = 12.0) ?
10. Calculate pH of :
- 10^{-1} M CH_3COOH
 - 10^{-3} M CH_3COOH
 - 10^{-6} M CH_3COOH
- Take $K_a = 2 \times 10^{-5}$
11. Calculate the pH of a solution which contains 100 mL of 0.1 M HCl and 9.9 mL of 1.0 M NaOH.
12. 0.25 M solution of pyridinium chloride $\text{C}_5\text{H}_6\text{N}^+\text{Cl}^-$ was found to have a pH of 2.699. What is K_b for pyridine, $\text{C}_5\text{H}_5\text{N}$?
13. What should be the hydrolysis constant of 0.01 M methylamine hydrochloride, if its K_b is 5×10^{-4} ?
14. Calculate the percent hydrolysis in a 0.06 M solution of KCN. [$K_a(\text{HCN}) = 6 \times 10^{-10}$]

15. Calculate the extent of hydrolysis of 0.005 M K_2CrO_4 . [$K_2 = 3.1 \times 10^{-7}$ for H_2CrO_4] (It is essentially strong for first ionization).
16. What volume of 0.2 M NH_4Cl solution should be added to 100 ml of 0.1 M NH_4OH solution to produce a buffer solution of pH = 8.7?
Given : pK_b of $\text{NH}_4\text{OH} = 4.7$; $\log 2 = 0.3$
17. Calculate the ratio of pH of a solution containing 1 mole of CH_3COONa + 1 mole of HCl per litre and of other solution containing 1 mole CH_3COONa + 1 mole of acetic acid per litre.
18. What % of the carbon in the H_2CO_3 , HCO_3^- . Buffer should be in the form of HCO_3^- so as to have a neutral solution ? ($K_a = 4 \times 10^{-7}$)
19. Calculate the pH at the equivalence point in the reaction of 22.0 mL of 0.10 M acetic acid, CH_3COOH , with 22.0 mL of 0.10 M NaOH .
20. Calculate the pH at the equivalence point in a titration of 50.0 mL of 0.40 M NH_3 with 0.40 M HCl .
21. The solubility of Ag_2CO_3 in water at 25 °C is 1×10^{-4} mole/litre. What is its solubility in 0.1 M Na_2CO_3 solution ? Assume no hydrolysis of CO_3^{2-} ion.
22. How much water is required to dissolve its 4 g of sodium sulphate at 323 K if its solubility at same temperature is 16% ?

